

TABLE OF CONTENTS

S-520 MISCELLANEOUS ARCHITECTURAL WORKS

	Page No.
1. SCOPE OF WORK	S 520-4
1.1 Doors	S 520-4
1.2 Windows	S 520-4
1.3 Louvers	S 520-4
1.4 Floor Finish	S 520-4
1.5 Built Up Roofing	S 520-4
1.6 Plumbing	S 520-5
1.7 Ceiling	S 520-5
1.8 Masonry	S 520-5
1.9 Carpentry Work	S 520-5
1.10 Building Appliances	S 520-5
2. CODES AND STANDARDS	S 520-5
3. DOORS	S 520-6
3.1 General	S 520-6
3.2 Swing Hollow Metal Doors	S 520-6
3.3 Rolling Steel Doors	S 520-10
3.4 Aluminium Doors	S 520-14
4. WINDOWS	S 520-15
4.1 General	S 520-15
4.2 Design Loads	S 520-15
4.3 Design Requirements	S 520-15
4.4 Material	S 520-16
4.5 Erection	S 520-18

5.	LOUVERS	S 520-18
5.1	General	S 520-18
5.2	Louvre Type	S 520-18
5.3	Design Load.....	S 520-18
5.4	Design Requirements	S 520-19
5.5	Material	S 520-20
5.6	Installation	S 520-21
6.	FLOOR FINISH	S 520-21
6.1	General	S 520-21
6.2	Types Of Floor Finish	S 520-22
7.	BUILT UP ROOFING.....	S 520-29
7.1	General	S 520-29
7.2	Materials	S 520-29
7.3	Permanent Roofing System	S 520-29
8.	PLUMBING	S 520-33
8.1	General	S 520-33
8.2	Drain System Criteria.....	S 520-33
8.3	Water Supply And Sanitary Equipment Work	S 520-35
8.4	Materials	S 520-52
8.5	Installation	S 520-53
8.6	Inspection And Testing	S 520-53
9.	CEILING WORK.....	S 520-56
9.1	General	S 520-56
9.2	Materials	S 520-56
9.3	Erection	S 520-58
9.4	Clean Up.....	S 520-58
10.	MASONRY WORK	S 520-59
10.1	General	S 520-59

10.2	Materials	S 520-59
10.3	Construction Of Hollow Concrete Block Work.....	S 520-62
10.4	Installation Of Ceramic Tile	S 520-63
11.	CARPENTRY WORK	S 520-63
11.1	General	S 520-63
11.2	Material.....	S 520-63
11.3	Workmanship.....	S 520-64
12.	BUILDING APPLIANCES & SPECIALTIES.....	S 520-66
12.1	General	S 520-66
12.2	Kitchen Unit.....	S 520-66
12.3	Steel Lockers.....	S 520-67

S 520 MISCELLANEOUS ARCHITECTURAL WORK

1. SCOPE OF WORK

Miscellaneous Architectural Work shall include the Bidder's designing, detailing, furnishing, installing inspection and testing of the following work covered by this specification and as applicable to the Bidder's building work included in Part 4, Subsection 1.2 "Summary of Work" and Technical Specification S-001 "General Design Requirement of Structure, Civil and Building Works". All materials and workmanship shall be neat and in accordance with the best modern practice and subject to prior approval of the Owner.

1.1 Doors

- Hollow metal doors.
- Rolling steel doors.
- Aluminium doors.

1.2 Windows

- Aluminium top hung.
- Aluminium horizontal projected.
- Aluminium fixed windows.

1.3 Louvers

- Fixed louvers.

1.4 Floor Finish

- Monolithic finish.
- Separate concrete floor finish.
- Ceramic tiles
- PVC tiles
- Glazed tiles

1.5 Built Up Roofing

1.6 Plumbing

- Floor drains.
- Roof drains.
- Plumbing fixtures.

1.7 Ceiling

- Louvered grid ceiling.
- Acoustic ceilings.

1.8 Masonry

- Hollow concrete block
- Thin set tiles

1.9 Carpentry Work

1.10 Building Appliances

- Kitchen unit.
- Steel lockers.

2. CODES AND STANDARDS

2.1 The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards “

2.2 The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards” are specifically applicable to the design, manufacture and testing of the work included in this Specification:

AA : Aluminium Association, U.S.A

AAMA : Architectural Aluminium Manufacturers Association, U.S.A

ACI : American Concrete Institute

AMCA : Air Movement and Control Association, Inc. U.S.A

ANSI : American National Standards Institute Inc.

ASTM : American Society for Testing and Materials

AWWA : American Water Work Association

LIA	:	Lead Industries Association, Inc. U.S.A
JIS	:	Japanese Industrial Standards
NAAMM	:	National Association of Architectural Metal Manufacturers U.S.A
NEMA	:	National Electrical Manufacturers Association, U.S.A
NFPA	:	National Fire Protection Association, U.S.A
SSPC	:	Steel Structures Painting Council, USA
UL	:	Underwriters Laboratories, U.S.A
N.I.5.	:	Peraturan Konstruksi Kayu Indonesia (Indonesian Regulation for Wooden Structure)
N.I.10.	:	Bata Merah Sebagai Bahan Bangunan (Indonesian Standard for Bricks)
IEC	:	International Electrotechnical Commission

3. DOORS

3.1 General

All doors shall be designed and provided by the Bidder and meet the following minimum requirements:

3.1.1 Design Load

3.1.1.1 Wind Load

Exterior doors shall be designed to withstand a minimum wind pressure of 150 kg/m² and interior doors 100 kg/m², uniformly applied over the entire area of the door. The maximum deflection of the door under wind pressure loading shall be 1/120 of the width span.

3.1.1.2 Operating Load

All exterior doors shall be designed to operate under a minimum of 50 kg/m² load uniformly applied over the entire area of the door.

3.2 Swing Hollow Metal Doors

3.2.1 Hollow Metal Door Usage

3.2.1.1 Class I Doors

Seamless hollow steel construction, with minimum 1.6 mm thickness face sheets and additional tubular reinforcement of head, hinged jamb and lock sets or push-pulls and prepared for surface hinges and through bolted to door, unless otherwise indicated.

Class I door shall be used for all exterior door locations. Class I doors shall also be used for interior doors that are exposed to heavy traffic areas, doors that are exposed to abusive usage or doors that are exposed to extreme difference in room pressure.

3.2.1.2 Class II Doors

Seamless hollow steel construction, with minimum 1.3 mm thickness face sheets and additional channel reinforcement at head and prepared for mortise hinges, unless otherwise indicated. Class II doors shall be used for all interior door locations.

3.2.2 Hollow Metal Door Type

The doors shall be flush type. Vision light shall be used when desired for a safety reason or required for a plant operating reason i.e. to view what is on the opposite side of the door when the doors is in the closed position.

3.2.3 Materials

3.2.3.1 Face sheets shall be roller-leveled or stretcher-leveled prime quality cold-rolled carbon steel.

3.2.3.2 Stiffeners, reinforcing and other accessories shall be prime quality carbon steel.

3.2.3.3 Glazing

- a. Glazing shall conform to the following applicable requirements: ANSI or JIS
- b. Normal door glass shall consist of minimum 6 mm thick clear polished wire reinforced glass.

- c. Glazing compound shall be in accordance with the manufacturer's recommendations.
- d. Setting blocks and spacer shims shall be made of neoprene rubber and shall be provided as recommended by glass manufacturer.

3.2.3.4 Door Frames

The frames for hollow metal doors shall be steel channels frame (ASTM A36 or A6063S of JIS H4100) or pressed metal frames (prime quality cold rolled carbon steel). Where a fire rated frame is required, the frame construction shall be used.

3.2.3.5 Hardware

Hardware for the door shall include locks, butts, closer, holders panic devices etc. the fasteners shall be type 304 stainless steel for exterior doors and brass type for interior doors. All hardware shall meet the Owner's prior approval.

3.2.4 Door Construction Details

3.2.4.1 General Provision

- a. All door shall be seamless hollow steel construction conforming to ANSI A123.1 or JIS G3101, with each door formed from two face sheets of steel. No seam shall occur on the door faces or sides. The top and bottom of the door shall be closed and the internal construction shall consist of steel rib stiffeners.
- b. All doors provided in the air conditioned room shall be insulated with fibrous glass or mineral wool of 48.000 kg per cubic meter density, having Underwriters Laboratories certification for the service.

3.2.4.2 Door Details

- a. Bottom clearance : Shall be maximum 3 mm between bottom of door and finish floor
- b. For doors with full mortise butts: Doors and frames shall be most mortised to receive butts.
- c. Bottom drips : Shall be provided on all exterior doors

- d. Astragals : Shall be provided on the active leaf of all double doors
- e. Door stop seals : Shall be provided on all doors.
- f. Glazing details : Shall be provided with fixed glass stop on the exterior side and removable glazing bead on the interior side of a door.

3.2.5 Door Treatment

3.2.5.1 Exterior Doors

- a. Both sides of face sheets, all surfaces of internal stiffeners, reinforcing, etc. and all other steel accessories forming and integral part of door shall be hot dip galvanized in accordance with applicable requirements of ASTM A123, A153, A385, A386 or JIS G3302 (Hot –dip zinc-coated steel sheet and coils) or other applicable standard Specifications for hot dip galvanizing of materials. The minimum weight of galvanizing shall be that for G90. Poorly galvanized work will be rejected.
- b. After fabrication of the door is completed, all exposed surfaces shall be properly cleaned and shall receive primer paint baked on at 150⁰C for not less than one half hour.

3.2.5.2 Interior Doors

- a. Inside surfaces of doors shall be treated, before fabrication, with a rust inhibitor, as approved by the Owner.
- b. Exposed surfaces of doors shall be treated with Bonderite after fabrication.
- c. Primer paint shall be applied and baked on at 150⁰C not less than one half hour.

3.3 Rolling Steel Doors

3.3.1 Curtain

3.3.1.1 Exterior Doors

Curtain for exterior doors shall be of interlocking slats, roll formed (not drawn) from not lighter than 1.27 mm hot-dip galvanized steel sheets of structural quality, conforming to the applicable requirements of ASTM A446 for grade A steel or JIS A4705 (Components of rolling fire door for building) or batter, with not less than G90 galvanizing. Galvanized surfaces shall be provided with phosphate coating for good paint adhesion (painting shall be also by the Bidder).

3.3.1.2 Interior Doors

Curtain for interior doors shall meet the same requirements as for exterior doors, except that slats shall be of manufacturer's standard gauge required for these doors.

3.3.1.3 Slats

Slats of flat exterior design shall be formed in easy curves without sharp bends, and shall be of the proper size and cross section to maintain the deflection within safe allowable limits. Assembly of slats shall be such that damaged slats can be removed and replaced easily.

3.3.1.4 Coiling

Curtain shall be coiled on a steel pipe of sufficient size to carry the door load with vertical deflection not to exceed 0.80 mm per 300 mm of opening width, and evenly balanced with helical springs contained in the pipe. All springs shall be anchored to the tension rod, and held in position by the same adjusting wheel, which shall be accessible from the outside.

3.3.1.5 Weather stripping

- a. Weather stripping shall be provided for all exterior doors, essentially continuous around each door, for an airtight and weather tight seal.

- b. Rubber weather stripping shall be provided on both sides of guides to reduce the infiltration of air around the ends of curtain. Rubber shall be not less than 1.60 mm thick. Weather stripping shall be attached to guides with continuous flat bars arranged to permit replacement of worn weather stripping.
- c. Weather stripping at door head shall be street rubber, not less than 1.60 mm thick and shall be attached to head with a continuous flat plat to permit replacement of worn weather stripping.

3.3.1.6 Bottom Bar

The curtain bottom bar shall be made of two structural steel angles, ASTM A36 or JIS, with hot dip galvanized finish.

3.3.1.7 Combination Safety Device and Weather Seal

- a. This combination safety device and weather seal at the bottom of curtain shall be manufacturer's standard, with electrical control to stop and reverse curtain travel automatically. Weather seal shall be extended full length of the door to provide an airtight and weather tight seal.
- b. If any rolling steel door in closed position rests on the top of railroad track rails which project above floor level, the combination safety device and weather seal shall be in three separate sections (from jambs to rails on both sides, and between rails) and shall extend to floor level. A cutout in door at each rail shall be provided, with the width of cutout equal to rail head width plus 50 mm and the height of cutout equal to rails height above floor plus 75 mm. Reinforcement of door shall be made horizontally across cutout sections, and loop type rubber astragals shall be provided projecting 75 mm below top of cutouts when door is in open position.

3.3.2 Accessories

3.3.2.1 Coil Brackets

Coil brackets shall be high test gray iron castings of uniform texture, or fabricated from steel plate not less than 6.35 mm thick. All surfaces shall be smooth and even.

3.3.2.2 Hood

Curtain coil shall be housed within a sheet metal hood formed from not lighter than 0.635 mm hot dip galvanized steel sheets, conforming to the applicable requirements of ASTM A525 for G90 coating class, or better. Galvanizing shall be treated for good paint adhesion as hereinbefore specified for the curtain.

Separate galvanized steel sheet metal with baked enamel finish closure pieces shall be provided as indicated or as required, to close the top of the hood tight against the adjacent wall or structure to provide a weather tight seal.

3.3.2.3 Guides

Guide shall be formed from structural steel shapes, ASTM A36 or equivalent, to provide slots of sufficient depth for retaining the curtain in guides. Continuous locking bar shall be provided at each guide for anchoring wind locks. Guides shall be hot-dip galvanized.

3.3.2.4 End locks

Malleable iron end locks shall be provided on each slat for interior doors.

3.3.2.5 Wind locks

Wind lock on each slat shall be provided regardless of width, for all exterior doors, and for interior doors where required. Wind lock capacity shall be as required for the wind load specified in Clause 3.1.1.

3.3.3 Electric motor operated rolling steel doors shall be used for large access openings. Rolling doors shall bear the appropriate U.L. label or approved equal when occurring in fire rated walls.

3.3.4 Operators

3.3.4.1 Mounting

Operator shall be bracket mounted on hood except operators for exterior mounted doors shall be wall mounted inside the building.

3.3.4.2 Gearing

Operator gearing shall be high efficiency worm gearing, running in an oil bath, complete with spring-set 380 or 220 Volt AC solenoid operated brake completely housed in an enclosure with a degree of protection of IP53 if located indoor and a degree of protection of IPW55 if located outdoor.

3.3.4.3 Motor and Controls

a. Motors

Motors shall be designed for outdoor operation, high starting torque, hoist type, 380 volt, three-phase, 50 Hz, squirrel-cage, of sufficient power to operate doors at an approximate speed of 0.3 m per second. For detailed requirements see Part 4, Technical Specification E-208.

b. Pushbutton Station

Unless otherwise indicated, each operator shall be equipped with a momentary contract heavy duty three-button push button station, marked “OPEN”-“CLOSE”-“STOP” and with limit switches, of type acceptable to the Owner, which will stop motor at limits of travel. Reversing magnetic contactor complete with 380/110 volt control transformer and thermal overload elements will be furnished by others. Pushbutton station shall be for wall mounting, unless otherwise indicated.

c. Limit Switches

Limit switch shall be heavy duty type, in IPW55 as defined by IEC 529 or NEMA Type 4 enclosures for outdoor operation, with provisions for terminating Owner’s conduit and conductor wires. Limit switches shall be rated for 0.5A inductive at 220 V DC.

d. Control circuits

Shall be suitable for 110 Volt AC operation.

e. Emergency Chain Operation

Emergency hand chain operator, which will not affect the timing of the limit switch, shall be provided to operate each door in case of power failure or removal of motor for inspection and servicing.

3.3.5 Installation

Rolling overhead steel doors and door frames shall be installed by an installer approved by manufacturer, in strict accordance with manufacturer's printed instruction.

3.4 Aluminum Doors

3.4.1 General

Aluminium wide stile entrance doors fully glazed and the aluminium door frames are only used as indicated in the Bidder's approved drawings.

3.4.2 Materials

3.4.2.1 Door, frames and Trims

- a. All door, frames and trims shall be extruded aluminium, alloy 6063 with anodized finish and the colour will be selected later by the Owner without any additional cost.
- b. Thickness of any component of framing shall be minimum 3 mm.

3.4.2.2 Fasteners

All fasteners shall be 245 aluminium alloy and cadmium plated brass or bronze and shall closely match the finish of the aluminium.

3.4.2.3 Glazing

All glazing shall be minimum 12 mm thick tempered safety glass.

3.4.2.4 Hardware

Hardware for the door shall include locks, butts, closers, holders, panic device, etc. of premium quality.

3.4.3 Erection

All doors and door frame shall be set plumb, level square and shall be continuously caulked to provide weather tight seal at all joints between frames and masonry or concrete with silicone rubber sealant per manufacturer's recommendation.

4. WINDOWS

4.1 General

All exterior windows shall be aluminium top hung windows having float glass glazing material and operated by a hand wheel except for the office areas shall be anodized aluminium, horizontal projected with insulating glass, commonly know as "Thermopane".

4.2 Design Loads

4.2.1 Wind Loads

Design wind loads shall be 150 kg/m².

4.2.2 Wind Load Design

Unless otherwise indicated, all windows and window frames in exterior walls, including mullions, connections, fasteners and other accessories shall be designed to withstand the specified wind loads over the entire area of each window without permanent distortion and without exceeding allowable unit stresses and allowable deflections. Allowable unit stresses shall conform to the applicable requirements of AA Specifications for Aluminium Structures. Allowable deflections shall be 20 mm maximum.

4.3 Design Requirements

4.3.1 Height And Length

Height and width of any hung window section shall be determined by lighting and ventilation requirements, but not to exceed 1.25 m in height and 2 m in width for a single window unless otherwise indicated.

4.3.2 Mullions

All mullions shall be of fixed type, unless otherwise indicated.

Maximum Horizontal Spacing of Window: Mullions 60 cm nominal.

4.3.3 Vents

Entire area of each window shall be operable and shall open out to an angle of 45 degrees to provide a free air opening of approximately 75%, except where windows are indicated to be fixed.

4.3.4 Window Components

The minimum thickness of any structural components of the window framing, vents etc. shall be not less than 3 mm.

4.4 Material

4.4.1 Window Framing And Hinge Members

Extruded aluminium, Alloy 6063, with T-temper as required for the service and with anodized finish.

4.4.2 Flashing

Aluminium alloy Alcad 3004 and not less than 3 mm thick.

4.4.3 Weatherstripping

Vinyl rubber bulb seal.

4.4.4 Fasteners

Shall be # 14 self-tapping screws made of Type 305 stainless steel, cadmium plated, with bonded combination stainless steel and neoprene washers. Spacing of fasteners shall be 30 cm O.C. maximum.

4.4.5 Glazing

4.4.5.1 General Provision

a. Seal

All glazing shall be sealed with vinyl rubber bulb seal all around, between the glazing and the sash, to obtain weather tightness.

b. Fastening

Fastening of glazing sheets shall be by continuous clamping around the edges of each sheet and no holes shall pierce the sheets. Clamping action shall maintain the sheets in place under a load equal to at least twice the design load, but shall also permit for proper expansion of the sheet.

For fastening panels to supports, fasten to vertical mullions of windows with aluminium cap mullion strip designed to permit expansion of panels and fasten to horizontal members of windows with Type 305 stainless steel self-tapping screws, cadmium plated, with bonded combination aluminium or stainless steel and neoprene washers.

c. Cleaning

All glazing sheets shall be cleaned of all labels, dirt, paint etc. and washed cleanly inside and outside, just prior to hanging of sash in window openings.

d. Broken Glass

The Bidder shall replace all glass broken during construction at no cost to Owner.

4.4.5.2 Type

a. Glass

Shall be minimum 6 mm thick Gray tinted polished float glass with heat reflective coating.

b. Insulating glass

Insulation shall be minimum 25 mm overall thickness which consist of 6 mm gray tinted polished float glass with heat reflective coating outboard and 6 mm clear polished float glass inboard with 13 mm air space between.

Insulating glass shall be applied for windows in air-conditioned offices control, rooms etc.

4.5 Erection

All windows shall be set plumb, level square and shall be continuously caulked to provide weather tight seal at all joints between frames and siding, masonry, concrete or girts with silicone rubber sealant as per manufacturer's recommendation.

5. LOUVERS

5.1 General

The location and the area of the louvers shall be determined by ventilation requirements.

5.2 Louvre Type

All louvers shall be fixed type

5.3 Design Load

5.3.1 Wind Loads

All exterior louvers shall be designed for a wind load of 150 kg/m².

5.3.2 Wind Load Design

Unless otherwise indicated, all louvers and louver frames in exterior walls, including connections, fasteners and other accessories, shall be designed to withstand the specified wind loads over the entire area of each louver, without permanent distortion and without exceeding allowable unit stresses and allowable deflections. Allowable deflections shall be 20 mm maximum.

5.4 Design Requirements

5.4.1 Free Area

The free area shall be 45 % minimum as determined by using the equation in AMCA Standard 500.

5.4.2 Pressure Drop

Both the intake and exhaust static pressure drops through a louver with an interior screen shall not exceed 3.8 mm wg (water gauge) at a velocity of 240 m/min through the free area when tested according to AMCA Standard 500, Figures 5.4 and 5.5 (wall mounted).

5.4.3 Water Penetration

Water penetration of a louver shall not exceed 60 and 250 cc water per m² of free area in 15 minutes, at free area velocities of 240 and 300 m/min, respectively. The water flow rates and length of time for each test shall be the minimum values specified I AMCA Standard 500.

5.4.4 Louver Size

Louvers and louvers frames shall be 15 cm in depth. The maximum louver width shall be limited to 1.75 m and maximum louver height shall be limited to 5 m for a single louver.

5.4.5 Screens

All louvers shall be provided with aluminium bird screens.

5.4.6 Assembly

Louvers shall be assembled by mechanical fasteners.

5.4.7 Louver Blades

Louvers shall have drainable blades.

5.4.8 Louver Mullions

Horizontal and vertical louver mullions shall be provided as required.

5.4.9 Mounting Angles

Mounting angles shall be continuous aluminium mounting angles as required. When using vinyl gaskets for sealing, the mounting angles shall be grooved for permanent insertion of seals.

5.5 Material

5.5.1 Louver Frames and Blades

Louver frames and blades shall be extruded 6063-TS aluminium alloy. The minimum thickness of louver frames and blades shall be 3 mm.

5.5.2 Screens

Screen frames shall be 6063-TS extruded aluminium with a minimum thickness of 3 mm, and shall be rewirable. The screens mounted on the interior or exterior faces of louvers shall be 19 mm mesh. Wire shall be galvanized steel with 3 mm diameter inter crimped at 6 m intervals.

5.5.3 Seals

Blade and jamb seals shall be vinyl

5.5.4 Fasteners

Fasteners used in the assembly of louvers shall be stainless steel.

5.5.5 Finishes

5.5.5.1 General Provisions

- a. Each louver shall have a fluoropolymer finish, with colour matching the adjacent wall.
- b. Louver mullions and exterior screens shall have the same the same finish and colour as the louver.
- c. Screen assembly shall be painted prior to installation to louver frames.

- d. Mounted angles and interior screens shall be mill finished.

5.5.5.2 Fluoropolymer Finish

This finish shall be a polyvinylidene fluoride coating containing a minimum of 75 % Kynar 500 resin. The coating shall be oven baked in accordance with the manufacturer's printed procedures.

5.6 Installation

5.6.1 The installation of louvers shall be in strict accordance with the louver manufacturer's printed instructions.

5.6.2 Louvre frames shall be continuously caulked externally and internally. An acceptable alternate to interior caulking is either extruded vinyl bulb gaskets or sealing tape.

6. FLOOR FINISH

6.1 General

6.1.1 Floor finishes shall be applied in the various locations based on the Finishing Schedule attached in S001.

6.1.2 Separate concrete floor finishes shall not be applied until all machinery and equipment have been in place.

6.1.3 Where floor drains are required in monolithic or separate finish, a 3 mm slope per 30 cm in the finish floor from high point of floor to floor drain shall be used.

6.2 Types of Floor Finish

6.2.1 Monolithic Finish

6.2.1.1 Requirements

- a. Monolithic finish is one produce on structural concrete slabs which are poured all at one time with one concrete mix. Monolithic finish shall be of the driest consistency possible to work with sawing motion of screed.
- b. Surface shall first be screeded and then be floated to a smooth, compact surface. Any excess water shall be removed during floating operation by covering surface with burlap or similar material and spreading of cement or a cement-sand mixture, directly on top of burlap. Spreading of cement or cement-sand mixture directly on top of concrete shall not be permitted.
- c. After floating, the surface of concrete shall be covered by floor hardener and shall be smoothened by steel trowel, slatelike surface. Final trowel finish shall be done after concrete is so hard that no mortar accumulates on trowel and a ringing sound is produced as trowel is drawn over surface. Finish surface shall not be marked off in areas or scored in any manner.

A floor surface hardener shall be applied in accordance with the manufacturer's instruction for permanently exposed floors.

6.2.1.2 Uses

- a. Structural concrete floor slabs where monolithic finish is indicated in the Finishing Schedule in S001.
- b. Floor areas where resilient tile finished (asphalt, rubber, vinyl, etc.) are indicated or as required in the Finishing Schedule in S001.

6.2.2 Separate Concrete Floor Finish (Topping Slab)

6.2.2.1 Requirements

- a. A separate concrete finish for the following areas shall be applied over the rough slab finish. The separate finish shall be steel trowel concrete finish similar to monolithic finish and shall have minimum 250 kg/cm² compressive strength at 28 days.
- b. Provide bonding adhesive for separate concrete floor finishes up through 10 cm thick.

- c. Material of aggregate, cement, reinforcing bar and water shall conform to the requirements of S-340.
- d. Field control test shall cover all material and concrete test required and shall be done in accordance with the requirements of S-340.
- e. Cure, seal and dustproof surfaces of monolithic and separate concrete floor finished immediately after finishing by applying one coat of membrane curing compound, resin base type (wax base not permitted) at the rate of approximately 7 mm³ per litre, conforming to the applicable requirements of ASTM C309, type 1.

6.2.2.2 Uses

- a. All areas where switchgear equipment installed shall be provided with 75 mm separate concrete finish reinforced with welded wire fabric to permit accurate setting of the base channels and provide a true level surface in these areas. The top of the rough concrete shall be set 75 mm below the nominal floor elevation.
- b. Where specifically required for setting of the Bidder's other construction equipment.

6.2.3 Raised Floor

6.2.3.1 Requirements

- a. The raised floor shall have self-supporting system and can be adjusted for height (nominal height is 500 mm).
- b. The raised floor shall have adequate strength to withstand uniform live load 1500 kg/cm² and 500 kg individual load measured at the central of a panel.
- c. The quality shall be of prime quality and unless otherwise required by the Owner, the size shall be 600 x 600 mm

6.2.4 Ceramic Tile

6.2.4.1 Requirements

- a. Ceramic tiles shall be of prime quality and tile colour, size and pattern shall be approved by the Owner.
- b. The tiles shall be soaked and bedded on a prepared screed, in a 1 : 3 by volume cement/sand mortar with thickness of 10 mm. the tiles shall be laid square and true with 3 mm wide joints grouted in matching cement mortar.
- c. Provide bonding adhesive as specified in Clause 6.2.2.1.b.
- d. The Bidder shall include all cutting and fitting tiles to abutments and protecting floor tiles after laying. The tiles shall be polished, washed and left clean on completion.

6.2.4.2 Uses

Unless otherwise specified in the Finishing Schedule attached in S001, the ceramic tile shall be applied for the following places.

- a. Office areas
- b. Kitchen and locker areas
- c. Hallways and stair landings
- d. Other areas unless otherwise indicated.

6.2.5 Glazed Tiles

6.2.5.1 Requirements

- a. Glazed tiles shall be prime quality and the tile colour, size and pattern shall be approved by the Owner.
- b. The tiles shall be installed per Clause 6.2.4.1

6.2.5.2 Uses

Unless otherwise specified in the Finishing Schedule attached in S001, the ceramic tile shall be applied for the following places.

- a. Toilet
- b. Shower rooms
- c. Locker areas.

6.2.6 Chemical Resistant Special Coating Work

6.2.6.1 Scope

- a. The Bidder shall arrange for the services of a qualified special coatings manufactures to provide coating system (s) for the areas and under the conditions specified. Such system (s) shall be guaranteed by the Bidder for the period of time and to the extent herein specified.
- b. The areas required shall be indicated on the Bidder's drawings including concrete surfaces of floors, trenches, curbs and embedded steel items etc. Unless otherwise shown on the Bidder's drawings, the chemical resistant special coating shall be applied for the following plant:
 - b.1 Water Treatment Plant
 - b.2 Waste Water Treatment Plant
 - b.3 Chlorination Plant.
- c. The work shall include the selection of the total coating systems, preparations of substrate surfaces, priming, coating application, preparation of each coating to accept next coating, for a multi-coating application and treatment of the finish surface. The Bidder shall propose the coating material for the Owner's approval. The Owner will select the colour of the coating.

6.2.6.2 Documentation to be Submitted

The Bidder shall submit the following minimal documents for the Owner's review before selection of coating:

- a. Generic description and identification numbers of the proposed materials, recommended dry film thickness of each coating, application and colour if different for each area.
- b. Preparation procedures for concrete and steel, application procedures including, but not limited to sketches of corner, concrete crack, drain, curb, equipment base, joint treatment and curing methods.
- c. Ambient environmental and safety conditions required, prior to, during and after application until curing is completed.
- d. Test and experience data indicating proposed system's past success in performing under similar conditions.
- e. Detailed repair procedures.

6.2.6.3 Warranty

- a. The Bidder shall guarantee the coating system (s) provided, for a period of 5 (five) years following the commercial operation.
- b. This guarantee shall cover any failure of the coating system (s) relative abrasion, cracking, blistering, delaminating and discontinuities which occur during exposure to the service conditions specified in Clause 6.2.6.4.
- c. Upon notice by the Owner of any failures the Bidder shall promptly visit the site, examine the failure and advise the methods of repairs, time when repair can be made and environmental conditions required for repairing the failure.
- d. Repairs under the guarantee shall be at no cost to the Owner and shall be performed promptly at the Owner's convenience.
- e. Such repair or removal and replacement shall be guaranteed for the remaining time of the original guarantee.
- f. If desired the Bidder will be allowed to periodically inspect the coating system as to continuation of specified service conditions and possible sign of imminent failure.

6.2.6.4 Service and Surface Conditions

- a. Chemical exposures
 - a.1 Floor surfaces shall be considered as subject to “splash and spillage”, trenches shall be considered as totally immersible areas, with trenches immersed 50 percent of the time. These conditions will prevail throughout the estimated service life of the coating (s).
 - a.2 Coated areas may be exposed to either acidic or caustic conditions based on the worst possible combinations of the fluids as determined by the Bidder.
- b. Abrasion exposure
 - b.1 Floor Surfaces: Constant foot traffic, occasional rubber wheeled fork lift traffic, empty and loaded.
 - b.2 Trenches: Slow flow of effluent.
- c. Temperature.
 - c.1 Floor surface: Operating temperature range varies from 20 °C to 50 °C.
 - c.2 Trenches and Sump Surfaces: Operating temperature range, 20°C to 50°C; temperature shock as determined by the Bidder for the effluent.
- d. Exiting Conditions of Surfaces.
 - d.1 Concrete Cracks: Concrete may have shrinkage cracks, the coating system shall accommodate, without cracking (failure) a 0.25 mm in width movement of the exiting cracks and the occurrence of new cracks in the concrete.
 - d.2 Concrete finish shall have a minimum age of 30 days prior to the application of the coating. Concrete shall be made using Type II cement and shall have minimum of 250 kg/cm² compressive strength at 28 days.
 - d.3 Steel : Surface may or may not have shop applied prime coats.

6.2.6.5 Work Conditions

- a. The Bidder shall protect existing equipment and surfaces which are not to be lined against coating deposits, dust and other detrimental factors associated with coating work.
- b. Coating shall be applied after completion of the equipment erection or as directed by the Owner..

6.2.6.6 Quality Control and Assurance

- a. Application of the coating system shall conform to the printed instructions of the coating manufacturer regarding environmental, preparation of surface, material mixing, material application, curing and repair.
- b. Material Information.
- c. Documentations indicating that the coating system (s) have been applied, cured and inspected per coating manufacturer's requirements shall be submitted to the Owner.
- d. Documentation indicating that coated surfaces in trenches have been spark-tested by the Bidder with voltage (s) required by the Bidder by Criteria shall be submitted to the Owner.

6.2.6.7 Acceptance Criteria

The surfaces coating system (s) shall meet the following criteria before acceptance by the Owner.

- a. Floor shall have evenly distributed nonskid surface.
- b. Trenches shall have smooth surface.
- c. The surfaces shall be free from blistering, delamination, crack and discontinuities.
- d. The coating system shall be of the required minimum and maximum dry film thicknesses acceptable by the Owner.
- e. Surfaces shall be uniform in texture, colour and appearance.

7. BUILT UP ROOFING

7.1 General

The built-up roofing over concrete/precast concrete slabs shall be comprised of alternate four layers of felt and four layers of bitumen to form flexible roof cover with sufficient strength to resist normal expansion and contraction forces, and a top cover layer.

7.2 Materials

7.2.1 Cant Strips

Fiberboard material, 10 cm high by 10 cm wide, bevelled at 45°.

7.2.2 Asphalt Primer

Solution of petroleum residual asphalt in petroleum solvent, per ASTM D41.

7.2.3 Vapor Retarder

Asphalt coated base sheet 1.8 kg/m² per ASTM D2626. for use only on concrete and precast roofs.

7.2.4 Bitumen (Asphalt)

Coal Tar (Type III, per ASTM D450).

7.2.5 Permanent Roof System

Coal-Tar-Saturated organic felt (0.65 kg/m²) per ASTM D227.

7.2.6 Roof Walkway Traffic Pads

Roof walkway traffic pads shall be ceramic tile.

7.2.7 Reglets, Metal Flashing and Counter Flashing

Shall be 0.47 mm thick stainless steel conforming to ASTM A240, type 304.

7.3 Permanent Roofing System

7.3.1 Asphalt Primer

7.3.1.1 The exiting concrete/precast concrete channel roof slabs shall be sealed with an asphalt primer. Apply the primer at a rate of 0.2 litre to 0.4 litre per m².

7.3.1.2 Applications and curing shall be in accordance with manufacturer's printed instructions.

7.3.2 Vapor Retarder

7.3.2.1 Provide single-ply asphalt coated base sheet on the roof area using asphalt bitumen by solid mopping at a minimum rate of 1.12 kg/m².

7.3.2.2 The vapor retarder shall be lapped with 10 cm over the preceding sheet and 15 cm at the end laps. Extend the vapor retarder 15 cm up all vertical surfaces.

7.3.2.3 The vapor retarder (coated base sheet) shall be manually broomed into the hot asphalt and laid without wrinkles, buckles or kinks and free from pockets and blisters.

7.3.2.4 The vapor retarder shall be applied so that the direction of water flow is not against the laps.

7.3.3 Perforated asphalt roofing felt with thin layer sand

7.3.3.1 Quality of material shall comply with JIS A6023 or equivalent.

7.3.3.2 The Bidder shall submit the sample to the Owner for approval.

7.3.4 Applications of Felts

7.3.4.1 Coal-tar bitumen shall be uniformly applied over the vapor retarder at the rate of 1.22kg/m². Into this coating of pitch, while hot, lay four layers of coal-tar saturated felts lapping each sheet 0.7 m over the preceding one and lapping the ends of the felts not less than 15 cm. End stripping or tapping of these laps will not be permitted. Each ply shall be uniformly mopped, using not less than 1.22 kg/m² bitumen. At no time shall the mopping extend more than 2.5 m ahead of the felt roll.

7.3.4.2 When beginning the membrane roofing installation, install the appropriate width starting strips of felt to achieve the required four-ply system.

7.3.4.3 Lay plies of felt in single fashion. Apply felts so that the direction of water flow is not against the laps of the plies.

7.3.4.4 Felts shall be manually broomed into the hot bitumen and laid without wrinkles, buckles or fish mouths so that the felts are free from pockets and blisters.

7.3.4.5 At all vertical surfaces extend the felts up 50 mm above the cant strip and mop.

7.3.5 Application of Insulation

7.3.5.1 Insulation material shall be confirmed to ASTM C534 or JIS A9511 (Preformed cellular plastics thermal insulation materials).

7.3.5.2 Insulation material shall be install between asphalt layer and cover concrete block or ceramic tile.

7.3.5.3 Sample of insulation material shall be submitted to the Owner for approval.

7.3.6 Composition of Waterproofing

7.3.6.1 Built-up asphalt waterproofing shall be constructed in accordance with the following Table:

Table S-520-1 Composition of Class A

Work Order	Flat	Rise
1	Asphalt primer (0.3 l/m ²)	Asphalt primer (0.2 l/m ²)
2	Perforated asphalt roofing felt (2.5 kg/m ²)	Asphalt roofing
3	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
4	Asphalt roofing	Asphalt roofing
5	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
6	Asphalt roofing	Asphalt roofing
7	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
8	Asphalt roofing	Asphalt roofing
9	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
10	Asphalt roofing	Asphalt (1.2 kg/m ²)
11	Asphalt (1.2 kg/m ²)	Insulation materials
12	Asphalt (1.2 kg/m ²)	Lean concrete with fiber material
13	Insulation materials	

Work Order	Flat	Rise
14	Sand	
15	Mortar	Mortar
16	Ceramic tile	Ceramic tile

Table S-520-2 Composition of Class B

Work Order	Flat	Rise
1	Asphalt primer (0.3 l/m ²)	Asphalt primer (0.2 l/m ²)
2	Asphalt roofing	Asphalt roofing
3	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
4	Asphalt roofing	Asphalt roofing
5	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
6	Asphalt roofing	Asphalt roofing
7	Asphalt (1.2 kg/m ²)	Asphalt (1.2 kg/m ²)
8	Mortar	Mortar
9	Glazed ceramic tile	Glazed ceramic tile

7.3.6.2 The class A shall be applied for the roof of the buildings.

7.3.6.3 The class B shall be applied for the lavatory and shower room.

7.3.6.4 The inclination of the base concrete shall be not more than 1/100.

7.3.7 Application Method

7.3.7.1 Asphalt primer shall be evenly sprayed over the base concrete. Asphalt compound shall then be evenly poured and spread over the asphalt primer. Compound heated to a temperature exceeding 280°C shall not be used.

7.3.7.2 In the case of Class-A with flat, perforated roofing shall be laid between the base concrete and asphalt compound.

7.3.7.3 Asphalt roofing and special roofing shall be flatly laid over each respective asphalt compound. The sides and ends of these sheets shall be provided with an overlap of at least 15 cm. The joints shall be completely water tight and not concentrated.

- 7.3.7.4** Prior to commencement of the work, the Bidder shall submit to the owner for approval the personal histories of the Waterproofing workers, who shall have 5 years experience or more. Care shall be taken for roofing work surrounding anchor bolts, parapets and roof drains to prevent any leakage.

8. PLUMBING

8.1 General

- 8.1.1** Plumbing work shall include the Bidder's designing, furnishing, installing, cleaning, painting, coating, inspection and testing of the following work specified hereinafter :

- a. Floor drains
- b. Roof drains
- c. Water supply and sanitary equipment work
- d. Plumbing fixtures and accessories

- 8.1.2** Piping from drainage and supply system shall not be routed over electrical equipment. If such piping must be routed over the electrical equipment, it shall be insulated.

8.2 Drain System Criteria

8.2.1 Floor Drains

- 8.2.1.1** Floor drains shall be provided at all concrete floors for normal wash down, drainage of any water accumulation from equipment drain or to drain water used in fire protection system.
- 8.2.1.2** All drain piping shall have a minimum slope of 3 mm per 300 mm with an accessible cleanout in each change of direction.
- 8.2.1.3** Sleeves are not required for floor drain piping.
- 8.2.1.4** All floor and equipment drainage shall flow through an oil separator before discharging to a sump, tank or exterior drain system.
- 8.2.1.5** In a monolithic finish, the top of drain shall be placed 10 mm below finished floor elevation with a drainage area of 40 to 50 m² per drain.

8.2.1.6 All floor drain grating shall be made of cast iron.

8.2.2 Roof Drains

8.2.2.1 Rainfall intensity: shall be maximum recorded in the area. The Bidder shall investigate the applicable rainfall intensity.

8.2.2.2 Roof drains for all major roofs shall be located in the center of every other column bay to a maximum of 15 m centers, or 800 m² of drainage area with a minimum size of 15 cm diameter. On smaller roofs 10 cm diameter drains may be used. Penthouse roofs and canopies may have 7 cm diameter drains.

8.2.2.3 The roof drainage down comer shall be discharged through downspout to the drainage ditch on the ground.

8.2.2.4 All roof drain shall be made of cast iron.

8.2.3 Material

8.2.3.1 Floor Drains

a. Base Slabs (except as noted otherwise)

a.1 All piping shall be of service weight cast iron coil pipes and fittings.

a.2 floor drains shall have cast iron body and strainers with caulked outlet and extensions as required. Strainers to meet finished floor level.

b. Above Base Slabs (except as noted otherwise).

b.1 Use “Victaulic type” couplings and fittings with galvanized steel pipe.

b.2 If floor drain system must be connected to sanitary system use cast iron screwed drainage fittings with galvanized steel pipe.

b.3 Floor drain shall be of cast iron body and grate with screwed bottom outlet.

8.2.3.2 Roof Drains

Hot dip galvanized butt-welded steel pipe with “Victaulic type” conforming to the following ASTM specifications.

- a. 250 mm dia. or smaller : ASTM A53 Schedule 40
- b. 300 mm dia. and larger : ASTM A53 Schedule 30

8.3 Water Supply And Sanitary Equipment Work

8.3.1 General

Water supply, sewage and sanitary work shall include the Bidder's designing, furnishing, cleaning, painting, coating, inspection and testing of the following work specified herein after:

- (1) Water supply equipment work
- (2) Sanitary equipment work

The Bidder shall submit shop drawings of the fabricated items to the Owner for approval. The shop drawings shall clearly show the details of fabrication, installation, dimensions, sizes, methods of anchoring and any other pertinent details required for satisfactory installation.

8.3.2 Equipment and Materials

8.3.2.1 Elevated Water Tank

- a. Type of elevated water tank

Elevated water tank shall be made of wind pressure proof and earthquake proof construction and consist of a sandwich construction panel with facing materials comprising fiberglass-reinforced polyester (F.R.P) made from unsaturated polyester resin and glass fiber of with foam plastic core.

- b. Materials

Unsaturated polyester resin shall be used for the facing material and shall be waterproof, weatherproof, and harmless to human health.

The fiberglass shall be made from non-alkali fiberglass as stipulated in JIS R4110.

The foam plastic core material shall be rigid and closed-cell.

Any fillers and coloring agents shall be harmless to human health.

c. Accessories

The elevated water tank shall be provided with adequate amount of the following accessories:

- c.1 Water inlet pipe connections (F.R.P.)
- c.2 Water outlet pipe connections (F.R.P.)
- c.3 Drainage pipe connections (F.R.P.)
- c.4 Connections for overflow hole and overflow pipe with insect screens (The insect screens shall be made of plastic and the connections shall be made of F.R.P.)
- c.5 Air passage and air vent with insect screens (The insect screens shall be made of plastic and the passage and vent shall be made of F.R.P.)
- c.6 Covered inspection opening (plastic)
- c.7 Reinforcing materials, supports and stands for elevated water tank
- c.8 Steel ladder
- c.9 Electrode mount
- c.10 Breakwater cover

8.3.2.2 Sanitary Equipment and Accessories

- a. All sanitary ware plumbing fixtures shall be of high quality and of Indonesian manufacturer, specified in Table S-520-1, through S-520-8 or approved equivalent.
- b. All fittings for sanitary wares shall be in accordance with JIS A 5514.
- c. All accessories and visible sanitary wares such as faucets, flush valves and flushing pipes shall be nickel chromium-plated.

8.3.2.3 Piping Materials and Pipe Fittings

- a. Pipe
 - a.1 Water supply pipes

The water supply pipe shall be of the steel pipe lined with rigid PVC in accordance with JWWA-K116.

a.2 Drainage and air vent pipes

The drainage and air vent shall be galvanized steel pipe in accordance with JIS G3442, rubber joints type cast iron pipe in accordance with JIS G5525 and HASS-210, rigid PVC pipe type VP in accordance with JIS K6741 and/or the centrifugal reinforced concrete pipe as specified in JIS A5372. The above pipes shall be applied for the drainage and air vent pipes.

b. Pipe Joints

b.1 Water supply pipe joints

The water supply pipe shall be joined using malleable cast iron flanged fittings coated with plastic in accordance with JIS B2301. The flange type pipe joints shall be made by welding the flange to the end surface of the steel made joints as stipulated in JIS B2211 or JIS B2212, and further by lining the internal surface with rigid PVC of the same specifications as that used for lining of the steel pipe.

b.2 Drainage and air vent pipe joints

The drainage and air vent pipes shall be joined using. Screwed type drainage pipe joint shall be in accordance with JIS B2303. Rigid PVC pipe joint shall be in accordance with JIS K6739.

c. Gate Valves

c.1 Gate valves with a nominal diameter of 50 mm or less shall be of bronze with threaded ends stipulated in JIS B2023 and rated at 10 kg/cm².

c.2 Gate valves with a nominal diameter of 65 mm or above shall be of cast iron outside screw type with flanges ends as stipulated in JIS B2044 and rated at 10 kg/cm².

d. Check Valves

- d.1 Check valves with a nominal diameter of 50 mm or less shall be of bronze, swing type with treaded ends as stipulated in JIS B2025 and rated at 10 kg/cm².
- d.2 Check valves with a nominal diameter of 65 mm or over shall be of cast iron, swing type with flanged ends as stipulated in JIS B2045.

e. Automatic Air Release Valves (Air Vent Valve)

Automatic air vent valves shall be of a float type with a function to automatically release air, and shall be actuated precisely. This valve shall be so designed that it withstands the maximum service pressure.

f. Ball Taps

The main parts of the ball taps shall be made of bronze, and the ball shall be brazed to a copper plate. In case the nominal diameter of ball is 50 mm or less, the ball tap shall be of screwed type, but in case the diameter of the ball is 65 mm or over, the ball tap shall be of a flanged type.

The ball tap shall be so designed that it will be actuated precisely without causing water hammer due to rapid opening or closing.

g. Flexible Joints

Flexible joints shall be of a bellows type, and shall have sufficient flexibility and resistance against pressure. The bellows and protective steel band shall be made of cold rolled stainless steel plate/sheet (SUS-304) as stipulated in JIS G4305.

The length of one flexible section shall be 400 mm.

h. Strainers

The strainers shall be of Y-type. In case the nominal diameter is 50 mm or less, the strainers shall be of a screwed type made of bronze, but in case the diameters is 65 mm or oven, the strainer shall be of a flanged type made of cast iron.

i. Pipe Washers

Pipe washers shall be made of nickel chromium coated brass or stainless steel.

j. Pipe Sleeves

Pipe sleeves shall be made of steel pipe or steel plate with a thickness of 0.4 mm or greater (0.7 mm or greater in case the nominal diameter exceeds 200 mm). However, non water proofed floor pipe sleeve to be used indoors shall be made of laminated cardboard.

k. Pipe Support Fittings

- k.1 Pipe supports shall be resistant to contraction and expansion, rolling, etc. of pipe , and be of a construction having sufficient bearing strength against load of pipe when liquid is contained inside. The materials to be used shall be in accordance with JIS G3101 (Rolled Steel for General Structure).

All steel fittings shall be finished by galvanizing.

- k.2 The insert shall have sufficient strength for supporting the pipe and shall have a construction suitable for connecting hangers, etc. All inserts shall be made of cast iron, pressformed malleable cast iron or steel plate.

l. Cementing/Bonding Materials

l.1 Thread sealing material

- Thread sealing tapes shall be in accordance with JIS K6885 (Unsintered Polytetrafluoroethylene Tapes for Thread Sealing (Raw Tapes)), and shall be hazardous to human health.
- Paste sealing agent shall not be affected by the liquid in pipe and shall consist of contents suitable for the intended purpose. In case the agent is used for sealing of piping for drinking water, it shall not be hazardous, it shall not be hazardous to health.

1.2 Packings

Packing shall be in accordance with JIS K6353 (Rubber Goods for Water Works Service) JIS R3453 (Compressed), etc. and be suitable for the quality of the water, water pressure, temperature, etc.

1.3 Caulking lead

Caulking lead shall, in principle, be Pig Lead.

1.4 Caulking hemp

Caulking hemp shall, in principle, be jute of # 130 single thread which, when tied in a bundle, has a diameter or about 25 mm.

8.3.2.4 Drainage Pipe Fittings

a. General

- a.1 Water sealing depth for traps shall be 50 mm or more, and the total effective area of the drainage hole for strainers shall not be less than the sectional area of the drainage pipe.
- a.2 The materials for drainage pipe fittings shall be Grade 2 or higher as stipulated in JIS G5501 (Grey Iron Castings) in case of cast iron materials. For brass casting, the materials shall be Grade 2 or higher as stipulated in JIS H5101 (Brass Castings)
- a.3 All iron castings shall be cold-painted having a synthetic resin finish containing refined bituminous material.
- a.4 The nickel chromium plated section of the drainage pipe fittings shall be equivalent to or higher than Class 1 of Grade 2 in JIS H8617 (Electroplated Coatings of Nickel and Chromium).

b. Floor Drain Trap

Floor drain trap shall be made of cast iron, and the strainer shall be nickel chromium plated brass. The floor drain trap for asphalt waterproofed floors shall be of a waterproofing type, but that for other floors shall be of an ordinary type.

c. Sink Trap

Sink traps shall be made of cast iron and shall be furnished with an inside basket made of brass or stainless steel (SUS 304). The strainer shall be made of brass having a nickel chromium plating finish.

d. Floor Clean-Out

Floor clean-out shall be of a screwed type made of brass having a nickel chromium plating finish. Floor clean-out for asphalt waterproofed floor shall be of a waterproofing type, but that for other types of floors shall be of an ordinary type.

e. Underfloor Clean-Out

Underfloor clean-out shall be of a screwed type made of brass.

f. Drainage Pipe Fittings

Drainage pipe fittings shall be made of brass having a nickel chromium plating finish, and the chain and stopper shall be made of stainless steel.

8.3.2.5 Pit and Pit Cover

a. General

a.1 The materials for iron castings shall be equivalent to or higher than Grade 3 as stipulated in JIS G5510 (Grey Iron Castings).

a.2 Iron castings shall be baked with refined tars in accordance with JIS K2473 (Processed Tars) to which more than 2 % linseed oil or drying oil is mixed, or shall be cold-painted with refined bituminous materials to which synthetic resin finish is added.

b. Invert Pit

- b.1 Invert pits shall be of concrete construction and all visible portions shall be finished by mortar coating. Moreover, the pit shall be furnished with a cover, and an invert applicable to the pipe diameter shall be provided on the bottom of the pit.
- b.2 The cover shall be made of cast iron with chain, and shall be of an odor-proof type able to withstand the weight of 2,500 kg.

c. Storage Pit

- c.1 Storage pit shall be of concrete construction and all visible portions shall be finished by mortar coating. Moreover, the pit shall be furnished with a cover.
- c.2 The cover shall be made of cast iron with chain, and shall be of an odor-proof type able to withstand the weight of 2,500 kg.

d. Valve Box

Valve boxes shall be of a circular type made of cast iron with a cover.

e. Garden House Valve Box

Garden house valve boxes shall be of a box type made of cast iron and furnished with a cover. The bottom of the box shall be so designed as to facilitate drainage of water.

f. Trench Covers and Frames

Frames and anchors shall be of all welded construction and of structural steel angles, covers shall be checkered plates.

Frame constructions shall be such that tops of frames and cover plates are finished flush with the floor.

Slots shall be provided for lifting. All sharp edges and burrs shall be removed from cover plates and exposed edges of frames.

Size of cover plates shall be designed to facilitate easy removal.

The thickness of checkered plate shall 6.0 mm, with stiffeners it required to limit deflections.

The material shall conform to JIS G3101-SS41, or equivalent.

g. Watertight Manhole Frames and Covers

Watertight manhole frames and covers shall be installed where indicated in the Bidder's drawings. The frames and covers shall be of heavy type with designed water pressure of 3 kg/cm² or more and of 700 mm diameter. Frames shall have flanges at base and top, rubber gaskets, and stainless steel cap with air vent valve.

8.3.3 Execution

8.3.3.1 Erection Works

a. Elevated Water Tank

- a.1 The location of the elevated water tank shall be arranged by the Bidder approved by the Owner.

The elevated water tank shall be firmly fixed with anchor bolts having sufficient strength so that the tank will not slide laterally nor move in any way due to horizontal seismic or wind loads.

- a.2 The elevated water tank shall be installed on a horizontal plane on the foundation having a steel base, and the foundation shall have an even bearing surface against the load. The elevated water tank and the steel base shall be fixed firmly with anchor bolts to withstand wind pressure and other loads.

- a.3 After installation, the elevated water tank shall be cleaned and washed with water. Following that the tank shall be sterilize by using sodium hypochlorite solution, etc.

- a.4 The piping related to the elevated water tank shall be so supported that weight of the piping will not be applied to the tank.

Flexible joints shall be provided for all respective connecting pipes to the water tank except for the drain pipe and air vent pipe.

b. Sanitary Ware, Accessories and Fittings

b.1 General

- In case wall hanging fittings are fixed on a concrete wall or brick wall, expansion bolts shall be used.
- In case a metal panel or lightweight steel framed board wall in set, steel plate and worked angle materials or hard wood patch shall be fixed to the sanitary ware in advance.
- In case a part of the sanitary ware is embedded in concrete, the portion of the sanitary ware that comes into contact with concrete or mortar shall be covered with asphalt having a minimum thickness of 3 mm.

However, the bottom contact surface of sanitary ware, such as that for standing type urinals, shall be filled with sand.

b.2 Stool (Water closet/WC)

- The upper end of the stool shall be set horizontally in place after deciding the precise setting position.

- Prior to connecting the stool to the drain lead pipe, the connecting end of the lead pipe shall be flared up to the diameter of the flange, and after inserting the non drying sealing materials between the external surface of the flange and connecting end, the stool shall be connected to the drain lead pipe by nuts fastened from above. The plate thickness of the external end of the flared lead pipe shall not be less than 2 mm. Moreover, the end of the flange for the lead pipe to be connected to the stool shall be supported sufficiently with hangers, etc. so that no load of drain pipe, etc. will act directly on the stool.

b.3 Urinal

- Urinals shall be set precisely in place without any misalignment.
- Connection of urinal with drain pipe shall be carried out in accordance with the procedures for connecting the stool in d2.
- The setting height of urinal shall be 530 mm from the floor surface to the upper front end of the urinal.

b.4 Wash basin and hand wash basin

- The bracket or back hangers shall be firmly set in place, and the wash basins shall be fixed carefully so that the upper surface of the basins will be kept horizontal. In order to eliminate leakage of water, heat-resistant non drying sealing materials shall be filled around drain holes of the basins and around openings between the drain pipe fittings.
- The settings height of the basins shall be 800 mm (approximate) from the floor surface to the upper front end of the basins.

b.5 Slop Sink

The trap shall be set in place without any misalignment, and the connection of slop sink to the drain pipe shall be carried out in accordance with the procedures for the above wash basins in (4).

b.6 Faucet

Water cocks shall be fixed firmly after precise centering by fully taking into account the convenience of use and harmony with the surrounding facilities.

A sufficient space for the spout shall be provided between the end of the spout of water cock and the flood level rim of the drain receptacle.

b.7 Mirror

The setting height to the upper end of the mirror shall be 1,800 mm from the floor surface.

8.3.3.2 Connection of Pipes

a. General

- a.1 All pipes shall be carefully cut at a right angle against the axial center of the pipes so as to avoid any deformation of the section, and the cut end shall be finished smoothly.
- a.2 All pipes shall be connected after removing any chips, dust and other foreign matter and after confirming that all such foreign matter has been completely removed from inside the pipes.
- a.3 In case piping work is temporarily suspended, all pipes shall be sufficiently protected to avoid the entry of foreign matter.

b. Water Supply Piping

- b.1 No water supply piping shall be connected according to an insertion system unless specified otherwise.
- b.2 Epoxy resin synthetic rubber rust inhibitors shall be sufficiently coated over the end face of threaded portions and the bottom of threaded portions of joints. All rust inhibitors to be applied to any drinking water piping shall not be hazardous to human health and shall not have any adverse effect on drinking water.

c. Drainage and Air Vent Piping

- c.1 Prior to connecting the galvanized steel pipes, the pipes shall be correctly threaded so that a slight clearance is provided between the end face of pipes and the recess of joint in order to obtain the tapered threaded pipe portion.

The steel pipes shall be screwed tightly into the joints

- c.2 All cast iron pipes shall be connected by using rubber rings. In this case, the pipes shall be inserted into the rubber rings until the end of spigot comes into contact with inserted into a position near the end of spigot in advance shall be inserted into the pipes carefully so that no twisting will occur in the opening between the socket and spigot. The , the rubber rings shall be set in close contact with the pipes while uniformly fastening the tap bolts and nuts using a junk ring.
- c.3 All rigid PVC pipe shall be carefully connected by a cold working method so that no ridges or abrupt diameter changes will occur between the connected pipes. The flow of water in piping shall be smooth and uninterrupted.
- c.4 All centrifugal reinforced concrete pipes shall be connected by using collars.

After both ends of the joined pipes have tight contact with each other at the central part of the collar and are carefully set in place to ensure proper alignment in the surrounding opening, stiff-consistency mortar shall be placed into the opening. In this case, all cement and water inside the pipes shall be completely removed. In case one end of a pipe is made into a socket from after connecting a collar to the end with mortar, mortar shall be placed about 10 mm away from the pipe end.

8.3.3.3 Piping

a. Water Supply Piping

a.1 General

- Prior to executive of piping work, detailed study shall be made on the relationship between other piping and equipment, and all positions for pipe laying shall be decided by taking into account the precise slope of the respective piping.

In the case where piping is laid inside a building, setting of the pipe support fittings and embedding of pipe sleeves shall be carried out without any delay and in accordance with progress of the work.

- Flange joints shall be inserted for all main piping at appropriate in order to ensure easy removal of such piping.
- In the case where piping is branched from the main piping, T-joints shall be used.
- Air release valves shall be provided at the dead air space portion in piping, and a mud discharge valve shall be provided at the portion where mud is collected. The size of the mud discharge valve shall be the diameter of the related pipe, however, the size of the valve shall be 25 mm as minimum.

- In case anti-sweat covering is not provided for piping, pipe washers shall be fixed to all portions of piping that penetrate through ceilings, floors, walls, etc. and is visible from the outside.
- Any openings between the piping that penetrates through fire service areas, etc. shall be filled with rock-wool heat insulation and other noncombustible approved material to maintenance the fire rating of the wall.
- In case where a pipe sleeve is used at a place requiring water tightness, lead caulking shall be provided in the opening between the sleeve and pipe.
- In the case where any steel pipe and similar materials are laid under the ground, such pipe shall be protected against corrosion by either a double coating of coal tars or a double winding of corrosion-proof vinyl tapes (JIS Z1901 “Protective Polyvinyl Chloride Tapes”).
- Any repair of steel pipe, cast iron pipe and lead pipe with caulking shall be prohibited.
- After completion of piping work, the inside of the piping shall be cleaned by water pressure during of hydrostatic testing of the piping.

a.2 Gradient

In the case of horizontally running pipe, the pipes shall have an upward slope and the down flow pipe shall have a downward slope. In principle, the gradient shall be 1/250.

a.3 Support Pitch

- The support pitch of the horizontally running pipes shall in principle, be in accordance with the values in the following table and, wherever necessary, all bent section and bracing sections shall be supported. In the case where steel pipes or similar pipes are supported with hangers, a steady rest shall be provided for each horizontally running pipe in avoid looseness of the pipe due to movement at the time of earthquake.
- A steady rest shall be provided for indoor vertical pipe at least at one spot on every floor.

MAXIMUM SUPPORT PITCH OF WATER SUPPLY PIPE

(Unit : m)

Nominal Pipe diameter (mm)	20 or less	25-40	50-80	100-80	200 or more
Maximum Pitch (m)	1.8	2.0	3.0	4.0	5.0

SUPPORT PITCH OF STEEL PIPE

Nominal Pipe diameter (mm)	20 or less	25-40	50	65-80	100 or more
Maximum Pitch (m)	1.0	1.0	1.0	1.0	2.0

b. Drainage and Air Vent Pipe

b1. General

- All Horizontally branched drainage pipes, etc. shall be combined nearly horizontally at an acute angle of less than 45° .
- In case bent lead pipe is used, the pipe shall be carefully fabricated so as not to impair its roundness, and no branch drainage pipe shall be connected to the bent section of the pipe.

- No waste water shall be discharged directly from the following equipment.

- * Cooling tower/s

- * Air conditioners

- * Elevated water tanks

- * Pumps

- Indirect drainage pipe shall be opened while maintaining a space of over twice the diameter or the pipe from the drain receptacle and flood level rim.

In case an obstacle is foreseen in providing the above opening, appropriate measures shall be taken for protection.

- The lowest portion of the vertical drainage pipe shall be fixed with a support stand where necessary.
- Air vent pipes shall be run out vertically or at an angle of less than 45° from the horizontally branched drainage pipe, and in no case shall the pipe be run out horizontally.
- In the case where air vent pipes on floors are connected to vertical air vent pipes, the said pipes shall be connected at the place over 150 mm from the flood level rim of the related equipment. Vertical air vent valves shall be connected to the stack vents in accordance to the stack vents in accordance with the above procedures.
- In case a pipe is buried, a pit shall be excavated from the lower end of the pipe to a depth of about 100mm in case the nominal diameter of the pipe is 300 mm or less, and about 150 mm in case the nominal diameter exceeds 300 mm. after unscreened crushed stone, unscreened gravel or pit sand has been laid and compacted, the pipe shall be laid. However, the portions where joints are provided shall be excavated further, if necessary

Initial back filling of pipe shall be carried out up to the center line of the pipe to ensure immovability, and after sufficient compaction, the pipe shall finally be buried.

Provisions other than those described in this paragraph shall be in accordance with general provisions in 4.3.4.1.

b.2 Gradient

The gradient of horizontally running drainage pipes laid inside a building shall be at least 1/100. All air vent pipes shall have an upward slope facing the vertical pipes and shall be free from reverse slope and unevenness.

b.3 Support Pitch

- The support pitch of steel pipes shall be in accordance with the applicable provisions.
- Horizontal cast iron pipes shall be supported at a pitch of less than 1.6 m, and the deformed fittings to be connected shall be supported at a pitch of less than 0.6 m respectively.

In case pipes are connected with rubber, each straight pipe and deformed fitting shall be supported at the connected portion. Vertical pipes shall be supported at least one spot on each floor.

- The support pitch of non metallic pipe (PVC pipes) shall, in principle, be as shown in the table below, and vertical pipes shall be supported at more than one spot on each floor.

SUPPORT PITCH OF VINYL PVC PIPE

(Unit : mm)

Nominal diameter	80 or less	100 or more
Maximum pitch	1.0	2.0

8.4 Materials

8.4.1 Plumbing fixtures and accessories shall conform to the requirement specified in Table S-512-1 through S-512-8 or approved equivalent.

8.4.2 All supply piping for plumbing fixtures shall be of copper piping with solder or threaded fitting.

8.4.3 All sewer piping shall be of vitrified salt glazed extra strength clay pipe per ASTM C700 and standard strength cast iron pipe per ASTM A74 with bell and spigot joints.

8.4.4 All supply piping for water shall be of hot-dip galvanized butt- welded steel pipe with threaded fittings.

8.5 Installation

8.5.1 All plumbing fixtures and accessories shall be installed in strict accordance with the manufacturer's printed instructions.

8.5.2 After installation is completed, the Bidder shall thoroughly clean all plumbing fixtures and accessories. Surface of all fixtures and accessories shall be free from stains markings, scratches and debris.

8.6 Inspection And Testing

8.6.1 The Bidder shall test all drainage system and plumbing fixtures and accessories after installation to ensure the proper operation as follows :

a. Water Supply Pipes :

A hydrostatic test of water supply pipes shall be carried out prior to coating and shielding and backfilling, or after completions of piping. The minimum pressure retention time shall be 30 minutes.

The test pressure of piping below elevated water tanks shall be twice the pressure equivalent to the static head.

b. Drainage Pipes

The water filling test of the drainage pipes shall be carried out prior to coating and backfilling, or after completion of piping.

The water passage test of sanitary drain pipes shall be carried out after sanitary wares, etc. have been attached.

The minimum water retention time in case of water filling test shall be 30 minutes or more and that in case of the water passage test shall be 15 minutes or more.

- 8.6.2** Any adjustments or repairs needed, shall be made by the Bidder's at his own expense and shall be completed to the satisfaction of the Owner.

Table S-520-3 Lavatories and Accessories

Item	Types/Description	Manufacturer/Model Numbers (for reference)
		Toto Indonesia
Lavatory	One piece white vitreous china lavatory.	L 511 V3
Supply & Drain Fitting	Combination fitting with aerator and pop-up drain.	TGL 511 – MPAN
Supply Pipe	9 mm wall supply, loose key stop, escutcheon, reducing, coupling, chromium plated	Standard
P-Trap	31 mm adjustable P-Trap, cleanout plug, tubing drain to wall, chromium plated	TGP
Mirrors	One over each lavatory	See Table S-520-7
Towel Rail	One at each lavatory, stainless steel rod and fittings	TS 113 W

Table S-520-4 Lavatories and Accessories

Item	Types/Description	Manufacturer/Model Numbers (for reference)
		Toto Indonesia
Water Closet	Indonesian Style	C97 V/A-97 KL & TV 150 NWV 12
	Floor mounted, siphon jet action, white vitreous china, elongated bowl, 37 mm top spud, 2 bolt caps.	C72/3/710 BRTL
Flush Valve	Chromium plated, with vacuum breaker, 2.5 cm screw driver angle stop and flush connections	Standard
Paper Holder	One each water closet. Forged brass, chromium plated, metal roller	S-20
Coat Hook	One each water closet. Forged brass, chromium plated.	TS 118 WS

Table S-520-5 Urinals and Accessories

Item	Types/Description	Manufacturer/Model Numbers (for reference)
		Toto Indonesia
Wall Hung Urinal	45 cm white vitreous china washout urinal with integral extended shields, flushing rim, strainer and trap with cleanout, and 5 cm back outlet.	U 57 M
Flush Valve	Chromium plated, with vacuum breakers, 18 mm screw driver angle stop and quiet flush action	Standard
Strainer	Beehive gird	Standard

Table S-520-6 Showers and Accessories

Types/Description	Manufacturer/Model Numbers (for reference)
	Toto Indonesia
Individual Type Showers:	TR 3418
Bent arm with universal ball joint, vandal proof, shower head, concealed piping	
One soap dist at each shower	
One towel pin at each shower	See Table S-520-7
Shower curtain and rod at each shower	

Table S-520-7 Miscellaneous Accessories

Types/Description	Manufacturer/Model Numbers (for reference)
	Toto Indonesia
Towel pins for lavatories and showers.	B-406
Soap Holder	S-156
Mirrors for lavatories and wash-fountains. Stainless steel angle frame, satin finish with concealed and theft-proof locking device. Vertical mount. 45 cm x 60 cm 62.5 cm x 120 cm Size as required	TS 119 AS5
Recess type shower curtain, stainless steel, 3.1 cm dia., satin finish, with duck shower curtain and stainless steel snap hooks.	
Water faucet adjacent to the water closet. 13 mm chromium plated for cold water only	T 23 B 13

9. CEILING WORK

9.1 General

9.1.1 This section covers acoustical ceiling and louvered ceiling, including supports and accessories.

9.1.2 Louvered grid ceiling shall be only used for control room area and other area required by the Owner.

9.1.3 Exposed tee suspended moisture resistant acoustic tile ceiling shall be used for in high moisture area such as toilet and shower rooms.

9.1.4 Exposed tee suspended acoustic tile ceiling shall be used for the other area such as office, kitchen, dining room etc.

9.2 Materials

9.2.1 Louvered Grid Ceiling

Louvered grid ceiling shall be of aluminium with aluminium lacquer finish and shall be square or rectangular with removable core.

9.2.2 Exposed Tee Suspended Moisture Resistant Acoustic Tile Ceiling

The following type and make with lay in panels, installed in exposed mill finish aluminium grid suspension system.

9.2.2.1 Suspension System : Exposed tee type, direct hung.

9.2.2.2 Tile : Moisture Resistant Acoustic Tile Laying in Panel with minimum 15.8mm thickness Travertine Design, washable matte white finish.

9.2.3 Exposed Tee Suspended Acoustical Tile Ceilings

One the following types and makes with lay in units, installed in exposed painted metal grid suspension system.

9.2.3.1 Suspension System

- a. Exposed tee system, direct hung, with interlocking cross tees and all required accessories.
- b. Exposed tee system, snap lock, direct hung, with interlocking beam cross tees and all required accessories.

9.2.3.2 Tile

- a. Incombustible mineral fibre with minimum 19mm thickness, shadow line panel glacier pattern, non-directional with aluminium foil back.
- b. Incombustible mineral fibre with minimum 19mm thickness, reveal edge, texture-tone pattern, non-directional with aluminium foil back.

9.2.4 Anchorage Of Hanger

9.2.4.1 Steel sub framing; Continuous hanger support steel sub framing, spaced 1.20 m on centers for acoustic tile ceiling are welded in place to the structural support.

9.2.4.2 Anchorage of hanger to concrete with sleeve type or self-drilling anchors, will not be permitted.

9.2.5 Tie Wire

Soft monel or 18-8 stainless steel, 1.587 mm

9.2.6 Hanger

All hangers shall be galvanized. Minimum size of galvanized annealed wire hangers shall be 2.778 mm for acoustic tile ceilings. The hangers size for each locations shall be as required by the weight of the suspended ceiling and spacing of the hangers. Provide extra hangers at each corner of the light fixtures and as required for the work, so that there are at least two additional hangers for each 1.20 m length of light in addition to the regular grid system hangers.

9.3 Erection

9.3.1 General Provision

9.3.1.1 Erection of all acoustical ceiling shall conform to the applicable requirements of ANSI Standard Specifications and to the requirements herein specified.

9.3.1.2 The ceiling grid and fixing system shall be designed and installed in a manner that will allow ready access for ease of maintenance.

9.3.2 Suspension System

Acoustical lay in type ceilings shall be unrestrained perimeter construction, providing freedom of movement at intersections with walls.

9.3.2.1 Hangers

Hangers shall be of ample length and shall be attached to the overhead constructions in such a way that the ceilings is securely and rigidly braced and supported so that it will not sag or become displaced as a result of either downward load or upward pressure. Attachments shall develop the full strength of each hanger.

9.3.2.2 Main Tees

Main tees shall be spaced not more than 60 cm apart with outer lines not more than 20 cm from walls and parallel thereto. Tees shall extend to within 25 mm of walls at ends and shall be continuous for their full length. Splices shall be made with manufacturer's standard splicers.

9.3.2.3 Furring

Cross tees shall be erected at right angles to main tees and spaced as required by the ceiling layouts. Cross tees shall be securely attached to the main tees at each crossing by special clips or by saddle tying.

9.4 Clean Up

After ceiling work has been completed, all rubbish shall be removed, surfaces of ceiling panels and trim shall be cleaned and floors shall be left clean.

10. MASONRY WORK

10.1 General

Masonry work shall include the following type of work as required with all associated reinforced bracing, anchors, ties and damp proof course:

10.1.1 Hollow concrete blocks masonry

10.1.2 Thin Set Ceramic Tile

10.2 Materials

10.2.1 Hollow Concrete Blocks

10.2.1.1 Concrete hollow blocks shall conform to JIS A 5406 or equivalent as approved by the Owner.

Concrete hollow blocks shall have no harmful distortion, cracks or defects

10.2.1.2 Cement, sand and aggregates to be used in the concrete hollow block shall be as specified in S-350 "Concrete Work". Coarse aggregate for blocks shall not be large, than one-fifth (1/5) of the narrowest width of hollow block section.

10.2.2 Thin Set Ceramic Tile

10.2.2.1 The size of ceramic tile shall be, proposed to the Owner for approval including standard quality, self spacing and glazed ceramic tile.

10.2.2.2 Color and pattern shall be approved by the Owner.

10.2.3 Uses

- a. Toilet room: full height of the wall
- b. Shower room: full height of the wall
- c. Kitchen: 1.75 m height of the wall
- d. Locker: full height of the wall

10.2.4 Mortar for Masonry

10.2.4.1 Mortar shall conform to the applicable requirements of ASTM C270 with a minimum compressive strength of 176 kg/cm², at 28 days and consist of the following materials :

- a. Sand : ASTM C144, use washed natural sand only
- b. Lime :
- b.1 Quick lime : ASTM C5
- b.2 Hydrated lime : ASTM C207, Type S
- c. Portland cement : ASTM C150, Type 1
- d. Mixing water : ASTM C270

10.2.4.2 Use either of following mixes (by volume) as best fits work and as required to produce best quality workmanship:

- a. Portland cement-lime-sand mix
 - a.1 Portland cement or approval equal
Type 1..... One part
 - a.2 Hydrated lime, Type S, or lime
Putty..... Not less than ½ part nor more than 1-1/3 parts
 - a.3 Masonry sand..... 2-1/4 to 3 times sum of volume of cement and lime.
- b. Masonry-cement-sand mix
 - b.1 Masonry cement..... One part
 - b.2 Masonry sand..... 3 to times volume of masonry cement.

10.2.4.3 Mixing

Mechanical batch mixer, in accordance with ASTM C270.

10.2.5 Reinforcing, Ties And Anchors

10.2.5.1 Continuous Reinforcing Ties

- a. Steel, heavy duty or extra heavy continuous truss type, hot dip galvanized after fabrication, shall conform to ASTM A82.
- b. Width of ties : 5 cm less than overall thickness of masonry construction.

10.2.5.2 Tie bar Anchors

- a. Type

Adjustable bar ties, sliding in bar anchors welded to supporting steel or sliding in bar or dovetail anchor slots securely fastened or anchored to concrete or masonry backing.

- b. Material for Tie bars and Bar Anchors

Steel, hot dip galvanized with minimum size of 5 mm diameter unless larger size is required.

- c. Shape and Make

Triangular, rectangular or oval ties bars, as required. Provide complete with the manufacture's standard for the construction

10.2.5.3 Corrugated Metal Ties (For double wythe wall)

Steel, hot dip galvanized, 79 mm thick, 2.5 cm wide by 17.5 cm long, corrugated.

10.2.5.4 Bar Reinforcing

Steel, ASTM A615, Grade 60, intermediate grade, deformed (except 6 mm diameter bars may be smooth).

10.2.6 Membrane Damp-proofing For Brick Work

2 plies of 0,67 kg/m² coal tar saturated felt.

10.3 Construction Of Hollow Concrete Block Work

10.3.1 Masonry work shall be plumb and straight about doors, windows, louvers and other openings, etc. The Bidder shall verify correct size and location of openings before masonry work is started.

10.3.2 The masonry wall shall be reinforced as follows:

10.3.2.1 Hollow Concrete Block Wall

- a. Unless otherwise specified reinforcing bars shall be of D 10 and shall be placed at three (3) block intervals horizontally and at one block intervals vertically.
- b. Intersection of the bars shall be securely tied with wire.
- c. The extra reinforcing bars to bars to be used for the perimeters of the opening shall be of D13.
- d. When the height of concrete block wall exceeds 30 times of the wall thickness, reinforced concrete bond beam shall be provided at intervals not exceeding 30 times the wall thickness.

10.3.3 Built in all concealed work as required for conduits, bushings, appliance and other built in work and for all openings, chases, recesses and slots.

10.3.4 Cover in all embedded portions of structural steel columns and beams with at least 13 mm thickness of sand cement mortar unless otherwise required.

10.3.5 Take special precautions in the erection of the masonry surfaces to maintain the necessary plumpness and alignment of masonry as required for thin set ceramic tile.

10.3.6 Provide membrane damp proofing for all exterior masonry resting on concrete foundations, floors, beams, curb etc.

- 10.3.7** All masonry wall surfaces shall be plastered with 15 mm thick mortar consisting of minimum 1 part of cement to 4 parts of sand by volume and shall be smooth finished.
- 10.3.8** The toilet and shower room walls shall be plastered with minimum 1 cement to two sand by volume (1:2) and shall include an approved water proof admixture.
- 10.3.9** Plaster wall shall be protected from draughts and from drying unevenly before it has set.

10.4 Installation Of Ceramic Tile

- 10.4.1** Install tile with the thin-set bed method using water resistant organic adhesives. Installation shall also conform to the applicable requirements of ANSI A108, "Standard Specifications for Installation of Ceramic Tile with Water Resistant Organic Adhesives".
- 10.4.2** Grout to be used for installation of tile work shall be subject to Owner's approval.
- 10.4.3** Cutting : Perform all cutting of glazed tile on the job with a power driven saw in such manner as to provide true and even edges.
- 10.4.4** Grout Joints : Slightly concave tooled.
- 10.4.5** Cove Bases : Provide cove bases for glazes side or sides of all walls.
- 10.4.6** Cleaning : Clean all tile surfaces with burlap as work progresses.

11. CARPENTRY WORK

11.1 General

Carpentry work shall include the supply, installation and finishing of all wooden doors, cabinets and other timber work as required and and/or as indicated on the approved Bidder's drawings.

11.2 Material

11.2.1 Lumber

- a. All carpentry work shall comply with the requirements of P.K.K.I."Peraturan Konstruksi Kayu Indonesia" 1961 –N.

- b. Lumber for doors and frames shall be teak class 1 and shall be well seasoned with moisture content not to exceed 15 % before preservative treatment.
- c. Lumber shall be free from knots, shakes, cracks and not or other defects which could affect its strength and durability.
- d. All lumber shall be treated with the approved preservative solution to preserve against moisture penetration, fungal decay, insect and vermin attack and to improve the dimension stability of the wood.
- e. All lumber shall be varnish finished.

11.2.2 Nails, Screws, Bolts And Nuts

All nails, screws, bolts and nuts shall be of sufficient size and strength and be adequately spaced to ensure that all lumber work and joinery work is constructed in the best possible manner.

11.2.3 Doors

- 11.2.3.1** All interior door shall be flush teak veneered on both sides of the door as indicated and varnish finished except in the toilet area where the doors shall have plastic vinyl lamination on the toilet side of the door.
- 11.2.3.2** All doors shall be 4 cm minimum overall thickness.
- 11.2.3.3** The Bidder shall supply and install all fixtures including door closers that shall meet the Owner's prior approval.

11.3 Workmanship

11.3.1 Delivery and Storage

The Bidder shall be responsible for seeing that finish materials are not delivered until the building and/or storage area is sufficiently dry so that woodwork will not be damaged by excessive changes in moisture content.

11.3.2 Protection

All finish materials shall be protected from weather and from damage of any sort during transit. After delivery at the site, all such materials shall be protected, as approved, to prevent damage from excessive changes in moisture content, or from any other damage. Any material, which is damaged, shall be replaced to the complete satisfaction of the Owner at Bidder's expense.

11.3.3 Joining For Joinery

11.3.3.1 All joint shall be tongue and groove, mortise and tenon or rabbeted and shall be glued, unless otherwise indicated, using glue suitable for the purpose of the complete work. Nails shall not be used in joints. Dowels shall be hardwood and at least two dowels shall be used in each joint. Completed joints shall be tight and so formed as to conceal shrinkage

11.3.3.2 Glued joints shall be used where provisions need not be made for shrinkage or other movements in the connections. All glued joints shall be cross tongued or otherwise reinforced. Surfaces to be glued shall have a good sawn or planed finish and be kept clean, free from grease, dirt, sawdust and any other contamination. Adhesives for joints shall be resin of a type approved by the Owner, with adequate pressure applied to ensure intimate contact whilst the adhesive is setting.

11.3.3.3 All joinery work should be left well sanded and finished smooth and ready receive a final varnish finish.

11.3.4 Nailing

Nailing shall be applied wherever possible, where not possible, nailing shall be so located and driven as not to be conspicuous on the surface. All exposed nails shall be carefully countersunk, and the holes shall be filled and the surface be finished to match surrounding wood finish.

11.3.5 Finish woodwork

11.3.5.1 Finish woodwork shall be installed plumb and level, straight and true and shall be carefully fitted and scribed (so as not mar or damage wood surfaces) to adjacent materials.

- 11.3.5.2** All exposed surfaces shall have a fine smooth finish and shall be free from machine or tool marks, abrasions, raised grain or other imperfections. Any imperfections found on completion of work shall be corrected, at the Bidder's expense, to the complete satisfaction of the Owner.

12. BUILDING APPLIANCES & SPECIALTIES

12.1 General

This shall include Bidder's furnishing and installing complete kitchen units and lockers as indicated on the Bidder's general arrangements drawings.

12.2 Kitchen Unit

- 12.2.1** The unit shall include but not be limited to the following including all accessories :

12.2.1.1 Electric range with two burners, and cooker hood

12.2.1.2 Electric refrigerator,

12.2.1.3 Double stainless steel sinks and countertop.

12.2.1.4 Upper cabinets.

- 12.2.2** Install kitchen unit complete with all plumbing connections, all electrical wiring, switches etc. integral with equipment and ready for plug in to electrical outlets and all conduit and outlet boxes as required. All electric appliances shall be suitable for operation on 220 VAC, single phase, 2 wire, 50 hertz.

12.2.2.1 Electric range with two burners.

12.2.2.2 Electric refrigerator

12.2.2.3 Double stainless steel sinks and countertop.

12.2.2.4 Upper cabinets.

- 12.2.3** Install kitchen unit complete with all plumbing connections, all electrical wiring, switches etc. integral with equipment and ready for plug in to electrical outlets and all conduit and outlet boxes as required. All electric appliances shall be suitable for operation on 220 VAC, single phase, 2 wire, 50 hertz.

12.2.4 The color of the appliances cabinets and counters will be selected by the Owner.

12.3 Steel Lockers

12.3.1 Metal lockers for lockers rooms shall be 46 cm by 53 cm deep by 65 cm high, double tiers with sloping top.

12.3.2 The shelves of all lockers shall be perforated and the upper and lower part of each locker door shall be fully louvered for proper ventilation.

12.3.3 Each locker shall have standard equipment including metal shelf, chrome plated handle, bolts and nuts and a combinations pad lock for each locker.

Doors shall be of heavy construction to resist warp of bending, each door shall be provided with a number plate, a name tag plate and means for padlock attachment.

Lockers doors shall be equipped with rubber bumpers.

12.3.4 Each locker shall be cleaned of all scale after fabrication, primed with a baked on primer and finished with baked on enamel of the color approved by the Owner.

(Last page of S-520)